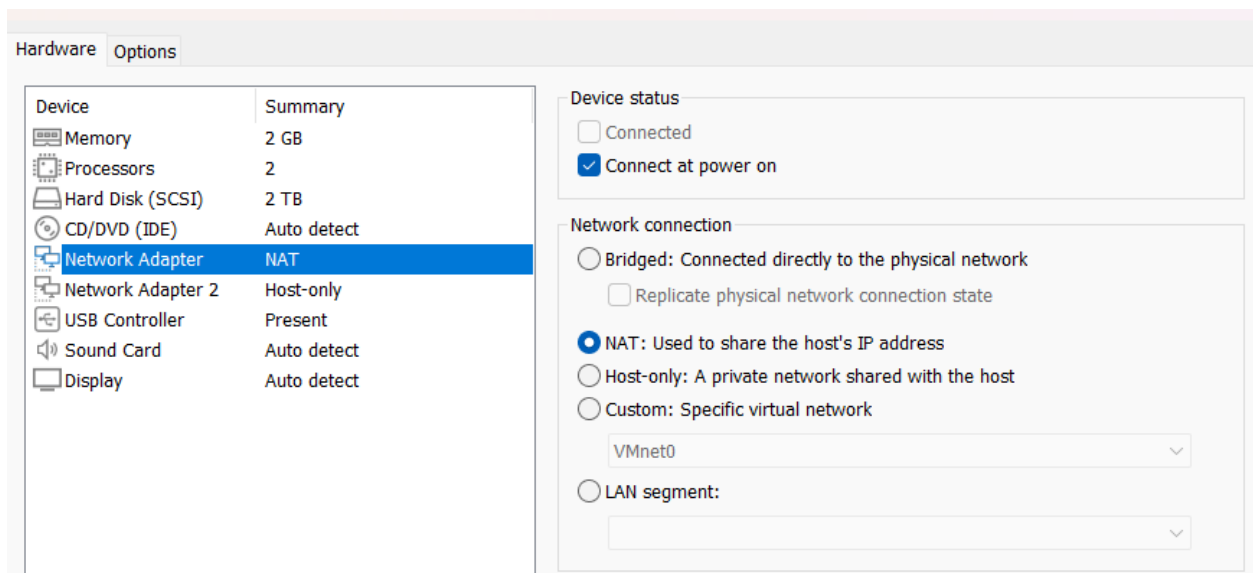
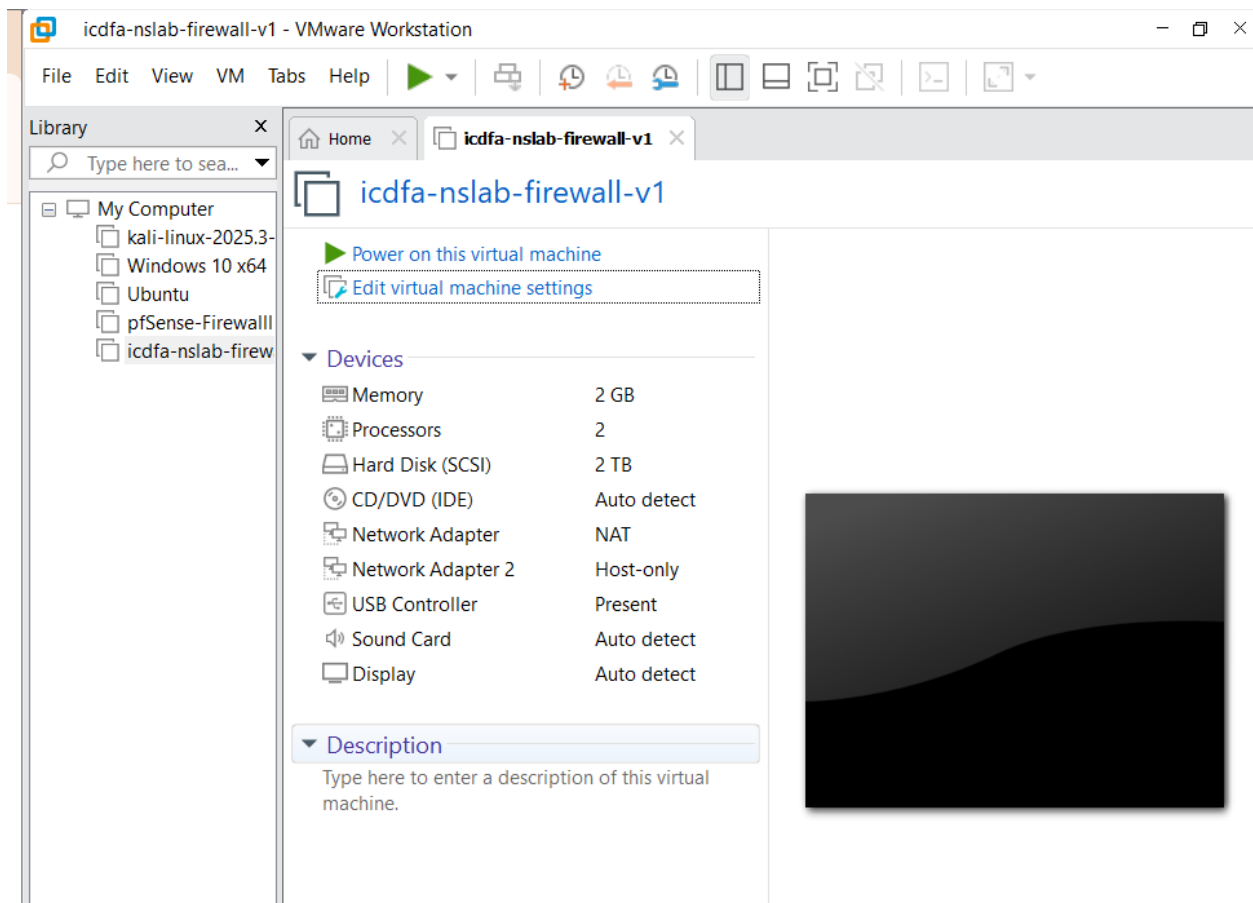




VMware network settings for icdfa-nslab-firewall-v1: Adapter 1 = NAT (WAN) and Adapter 2 = Host-only VMnet1 (LAN)."

```
Starting Dnsmasq...done.  
Starting Unbound DNS...done.
```

```
*** icdfa-nslab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***
```

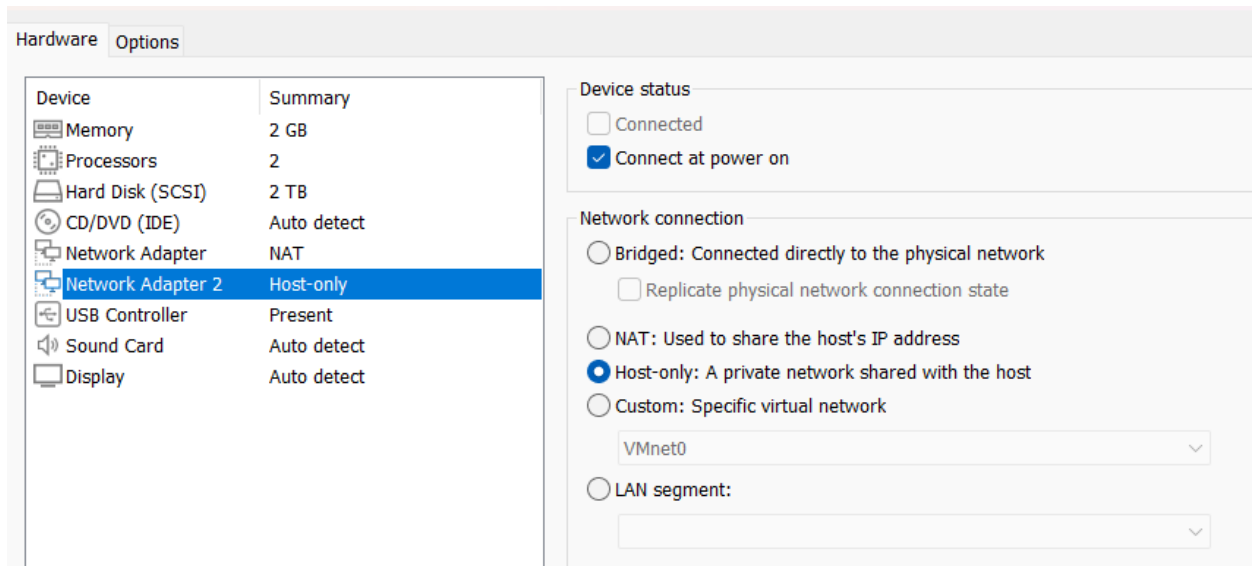
```
DMZ (em2)      -> v4: 10.10.20.1/24  
LAN (em1)      -> v4: 10.10.10.1/24  
WAN (em0)      -> v4/DHCP4: 192.168.128.137/24
```

```
HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22  
          82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90  
SSH:   SHA256 K10LtdmW484dUmuPy37tD7l4RA0De5LKjSU/+bKB1Tw (ECDSA)  
SSH:   SHA256 5wvyP8AD885sP0fHz7CgfNStFMhPwXXX42x23cBxco4 (ED25519)  
SSH:   SHA256 6e/0XrjSdfL/Hsy8atZw0IPQGsb1P6zFfuJ/FZ50d+A (RSA)
```

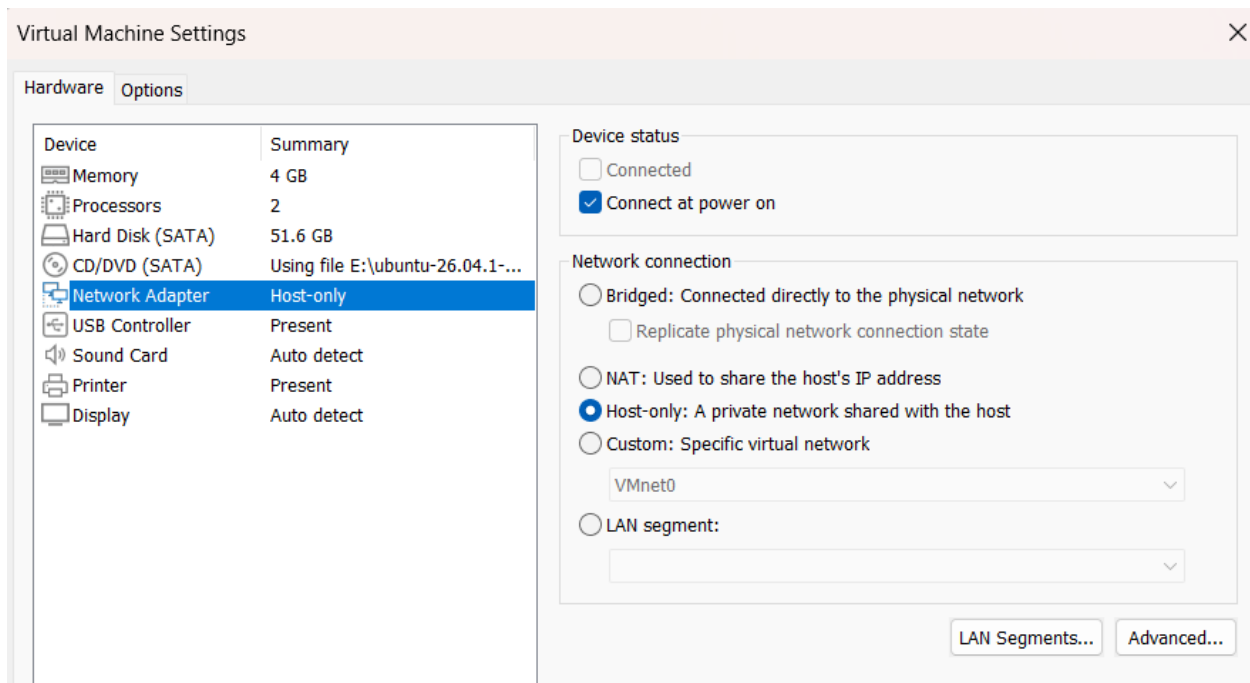
- | | |
|------------------------------|-------------------------|
| 0) Logout | 7) Ping host |
| 1) Assign interfaces | 8) Shell |
| 2) Set interface IP address | 9) pfTop |
| 3) Reset the root password | 10) Firewall log |
| 4) Reset to factory defaults | 11) Reload all services |
| 5) Power off system | 12) Update from console |
| 6) Reboot system | 13) Restore a backup |

```
Enter an option: █
```

OPNsense console for icdfa-nslab-firewall-v1 showing LAN (10.10.10.1/24 static) and WAN (192.168.128.137/24 via DHCP from VMware NAT)."



VMware network settings for icdfa-nslab-client-v1: Adapter 1 = Host-only VMnet1 (same protected LAN as firewall)."



Client VM (icdfa-nslab-client-v1) network settings: Memory 4 GB, 2 processors, 51.6 GB SATA disk, Ubuntu 26.04.1 ISO attached for recovery, Network Adapter set to Host-only on VMnet0 (same isolated LAN as the firewall)."

```
Internet:
Destination      Gateway          Flags           Netif  Expire
10.10.10.0/24    link#2          U               em1
10.10.10.1      link#4          UHS            lo0
10.10.20.0/24    link#3          U               em2
10.10.20.1      link#4          UHS            lo0
127.0.0.1       link#4          UH             lo0
192.168.128.0/24 link#1          U              em0
192.168.128.137 link#4          UHS            lo0

Internet6:
Destination      Gateway          Flags           Ne
tif Expire
::1              link#4          UHS
lo0
fe80::%lo0/64    link#4          U
lo0
fe80::1%lo0      link#4          UHS
lo0
root@icdfa-nslab-firewall-v1:~ # route add default 192.168.128.2
add net default: gateway 192.168.128.2
root@icdfa-nslab-firewall-v1:~ # netstat -rn | grep default
default          192.168.128.2    UGS             em0
root@icdfa-nslab-firewall-v1:~ #
```

```

root@icdfa-nslab-firewall-v1:~ # ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=128 time=38.849 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=38.018 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=38.176 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=46.531 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 38.018/40.394/46.531/3.557 ms
root@icdfa-nslab-firewall-v1:~ # █

```

```

root@icdfa-nslab-firewall-v1:~ # ping -c 4 google.com
PING google.com (142.250.200.174): 56 data bytes
64 bytes from 142.250.200.174: icmp_seq=0 ttl=128 time=54.353 ms
64 bytes from 142.250.200.174: icmp_seq=1 ttl=128 time=43.699 ms
64 bytes from 142.250.200.174: icmp_seq=2 ttl=128 time=42.491 ms
64 bytes from 142.250.200.174: icmp_seq=3 ttl=128 time=41.493 ms

--- google.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 41.493/45.509/54.353/5.165 ms
root@icdfa-nslab-firewall-v1:~ # █

```

```

round-trip min/avg/max/stddev = 41.493/45.509/54.353/5.165 ms
root@icdfa-nslab-firewall-v1:~ # curl -I https://opnsense.org
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 08:52:16 GMT
Server: Apache
Link: <https://opnsense.org/wp-json/>; rel="https://api.w.org/", <https://opnsense.org/wp-json/wp/v2/pages/199>; rel="alternate"; title="JSON"; type="application/json", <https://opnsense.org/>; rel=shortlink
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Type: text/html; charset=UTF-8

root@icdfa-nslab-firewall-v1:~ # █

```

"Firewall WAN connectivity tests after installing default route: (1) ping -c 4 google.com — 4/4 packets received with 0% loss, confirming ICMP + DNS resolution via WAN; (2) curl -I https://opnsense.org — HTTP/1.1 200 OK with HSTS security headers, confirming TCP/TLS/HTTPS connectivity through NAT."

```
root@icdfa-nslab-firewall-v1:~ # ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=128 time=41.001 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=39.439 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=40.359 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=39.269 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 39.269/40.017/41.001/0.703 ms
root@icdfa-nslab-firewall-v1:~ #
```

```
0 packets dropped by kernel
root@icdfa-nslab-firewall-v1:~ # ping -c 4 192.168.128.2
PING 192.168.128.2 (192.168.128.2): 56 data bytes
64 bytes from 192.168.128.2: icmp_seq=0 ttl=128 time=0.662 ms

64 bytes from 192.168.128.2: icmp_seq=1 ttl=128 time=0.564 ms
64 bytes from 192.168.128.2: icmp_seq=2 ttl=128 time=0.471 ms
64 bytes from 192.168.128.2: icmp_seq=3 ttl=128 time=0.572 ms

--- 192.168.128.2 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.471/0.567/0.662/0.068 ms
root@icdfa-nslab-firewall-v1:~ #
root@icdfa-nslab-firewall-v1:~ #
```

```
10:02:09.408890 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:10.291727 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:11.288737 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:16.940680 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:17.792328 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:18.794713 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:19.949844 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:20.781909 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10:02:21.783250 ARP, Request who-has 192.168.128.2 tell 192.168.128.1, length 46
10 packets captured
60 packets received by filter
0 packets dropped by kernel
root@icdfa-nslab-firewall-v1:~ # arp -d 192.168.128.2
192.168.128.2 (192.168.128.2) deleted
root@icdfa-nslab-firewall-v1:~ # ping -c 2 192.1668.128.22
ping: cannot resolve 192.1668.128.22: Name does not resolve
```

```
root@icdfa-nslab-firewall-v1:~ # host google.com
google.com has address 142.250.200.174
google.com has IPv6 address 2a00:1450:4019:80f::200e
google.com mail is handled by 10 smtp.google.com.
root@icdfa-nslab-firewall-v1:~ # fg
tcpdump -i em0 -n -c 10 udp port 53
10:04:20.077672 IP 192.168.128.137.44467 > 216.239.34.10.53: 50580% [1au] MX? go
ogle.com. (39)
10:04:20.243909 IP 216.239.34.10.53 > 192.168.128.137.44467: 50580*- 1/0/10 MX s
mtp.google.com. 10 (252)
10:04:20.244946 IP 192.168.128.137.36261 > 216.239.36.10.53: 56942% [1au] MX? go
ogle.com. (39)
10:04:20.394430 IP 216.239.36.10.53 > 192.168.128.137.36261: 56942*- 1/0/10 MX s
mtp.google.com. 10 (252)
```

"tcpdump capture on OPNsense WAN interface (em0) showing: (1) ARP request 'who-has 192.168.128.2' from the firewall, resolving the default gateway; (2) ICMP echo request/reply between 192.168.128.137 and 8.8.8.8 — 4/4 packets, 0% loss, ~40ms RTT; (3) DNS MX query for [google.com](https://www.google.com) sent to 216.239.34.10:53 with the corresponding response. These packets verify ARP resolution, ICMP reachability, and DNS name resolution at the packet level."

E7 — Why does the client VM use 10.10.10.1 as its default gateway?

"The client VM uses 10.10.10.1 as its default gateway because this is the LAN interface of the OPNsense firewall (confirmed in the firewall console: LAN (em1) = 10.10.10.1/24). When the client sends a packet to any destination outside its own subnet (10.10.10.0/24), the packet is forwarded to 10.10.10.1 because that's the only way out of the local network. OPNsense then applies Network Address Translation (NAT) and routes the packet via its WAN interface (192.168.128.137/24) using the default route 192.168.128.2, which was verified in the firewall's routing table: default 192.168.128.2 UGS em0. Without this default gateway, the client would only be able to reach hosts within its own subnet not the internet. This routing behaviour was demonstrated in the packet captures (E6) showing ARP resolution for the gateway and ICMP/DNS traffic exiting via the WAN."